

The French Model for Cooperative Semiconductor Research

Lessons from CEA-Leti

Sujai Shivakumar, Charles Wessner, and Thomas Howell

Leading advanced economies are showing a remarkable level of interest in securing the future of their semiconductor industries, driven by a growing appreciation of the sector's economic and strategic importance as well as a sharp realization of the strategic vulnerabilities inherent in its global supply chain. Several countries have announced new policies and programs to support the industry and the research that underpins it. Many of these involve substantial financial support in the form of direct subsidies and generous tax incentives. While there is a strong national and regional flavor to these efforts, there is also a growing realization that the success of national programs may well hinge on international cooperation, especially in research and training. Indeed, the companies themselves—however competitive in the marketplace—are often the first to emphasize the need for cooperation, especially in pre-competitive research.

Importance of Cooperative Research Organizations

Recognition of this reality has focused attention on the importance of the cooperative research organizations that bring together companies, university researchers, and their own highly trained staff to develop the technologies necessary to continue the semiconductor industry's unparalleled growth. The U.S. **CHIPS and Science Act**, the **European Chips Act**, and the **Rapidus consortium** in Japan all seek to strengthen and grow the role of these key research organizations. Leading centers include the Interuniversity Microelectronics Centre (IMEC) in Flanders, Belgium; SUNY Polytechnic Institute and its College of Nanoscale Science and Engineering (CNSE) in New York; and the Alternative Energies and Atomic Energy Commission's (CEA) "Leti" lab in Grenoble, France. Each of these institutions plays

a unique and valuable role in the international network of research and cooperation that underpins the global semiconductor industry.

Due to the central role of these institutions, it is important to understand their operations, organization, sources and scale of funding, and operational approach (i.e., membership and procedures), as well as the unique assets that each brings to the advancement of an industry that has transformed the world. Understanding these organizations' operations and roles in the global semiconductor network offers important lessons for the establishment of the National Semiconductor Technology Center (NSTC) called for in the CHIPS Act. With that goal in mind, this article reviews the Grenoble-based microelectronics research center known as CEA-Leti.

The Role of CEA-Leti

CEA-Leti is one of the world's leading nonprofit semiconductor research organizations, serving an intermediary role between basic research and industry applications. Other such organizations in Europe are Belgium's IMEC, Finland's VTT Technical Research Centre, Germany's Fraunhofer Society, and Spain's Tecnalia. CEA-Leti is part of a larger government organization, the French Alternative Energies and Atomic Energy Commission (CEA). Jean-René Lèqueypes, the deputy director of CEA-Leti, **says of his organization** that "we are an applied research laboratory, making the link with fundamental research. We collaborate closely with the fundamental research division within CEA, as well as with other research institutes outside of CEA, to explore novel paths for research. We take pure research and transfer it to industry in good condition."

CEA-Leti **represents** a French "innovative model combining a strong link to academia with an equally strong collaboration with industrial partners." The most powerful expression of this approach is **Minattec**, CEA-Leti's innovation campus in Grenoble, "where education, research and industry are meeting in a single location for enhancing the innovation through cross-fertilization of various micro- and nanotechnology fields and where CEA-LETI plays a key role in applied research."

CEA-Leti's single-minded focus on applied research reflects an extraordinary decades-long effort by French policymakers to restructure a sclerotic innovation system to restore their country's role as a global innovation powerhouse. This substantial effort is one that **some view** as demanding "far-reaching socio-cultural change, as seen in England at the time of the Industrial Revolution and Japan in the Meiji era." Whether France will ultimately succeed is unclear, but there has been sufficient progress that some observers suggest **the country** may be emerging as an innovation model for Europe—and potentially even the **world**.

CEA-Leti's single-minded focus on applied research reflects an extraordinary decades-long effort by French policymakers to restructure a sclerotic innovation system to restore their country's role as a global innovation powerhouse.

France's Overhaul of Its Innovation System

While France has **long exemplified** the European paradox of tending to be strong in basic research and weak in applied research, CEA-Leti is widely acknowledged to be a superb applied research organization and thus a powerful demonstration of how France is making progress in changing that dynamic. Current French innovation strategy, which has been implemented in phases beginning in the early 1980s, **emphasizes** the need to bridge a long-standing and deeply embedded gap between the national research base and industry. The effort involves nothing less than changing the country's innovation culture, a decades-long challenge faced not only by France but also by leading science research countries such as the United Kingdom and Canada. France has had to overcome several obstacles:

- **A focus on basic research:** The post-World War II French research base was dominated by the government's Paris-based National Center for Scientific Research (CNRS), the largest basic research organization in Europe. CNRS scientists traditionally accounted for an impressive array of Nobel prizes but were often **relatively uninterested** in applied research and technology development.
- **A dearth of technical skills:** The French education system was not producing enough qualified graduates with the skills needed by French industry and the national research base. Most of the best students at the *grandes écoles*, the elite universities that trained the country's leaders, showed **little interest** in doing scientific research or pursuing doctorates in the sciences. As **a 1986 report** by the Organization for Economic Cooperation and Development (OECD) found, "Although French society has always attached a great deal of importance to education. . . , this investment in education has for a very long time neglected the technical side, resulting in a marked deficiency in the technical culture of all sections of the population."
- **The crowding out of small innovators:** During the so-called *Trente Glorieuses* (30 "glorious" years of economic expansion between 1945 and 1975), certain national champion enterprises absorbed most of the public funds supporting applied research. These *grandes programmes*—large-scale government research and development (R&D) projects—led to major and lasting technological successes for some favored sectors such as highspeed rail, atomic energy, and, later, telecommunications and aviation but starved small and medium-sized enterprises (SMEs) **and** "other sectors of even the most basic support for technology innovation." In particular, the **"lack of collaboration between** start-ups and industrial groups in France . . . [has been] fatal for the quality of the French entrepreneurial ecosystem . . . [with the result that] the number of start-ups that have become 'number one' in their market is small."
- **Programmatic redundancy:** Successive French innovation policies tended to create new programs alongside existing ones. Instead of weeding out or consolidating unsuccessful programs, **the approach** was "often to create a new measure with more or less the same objectives but with a larger budget and/or a larger coverage," frequently leading to overlapping roles and duplication of efforts.

Recognizing these problems, President François Mitterrand initiated serious reform of French innovation policy. Beginning in **1982**, he oversaw the enactment of legislation establishing a special fund to promote innovation (emphasizing SMEs), introducing R&D tax credits, and incentivizing collaboration between public research organizations and industry. As the junior minister for industry

and foreign trade in the early 1990s, Dominique Strauss-Kahn made numerous visits to Silicon Valley, meeting and brainstorming with U.S. high-tech executives. Indicative of a radical rethinking of the French approach to innovation, in 1992 Strauss-Kahn **asked**:

Of what use is innovation if we do not have the techniques to industrialize and commercialize the products that result from this innovation? It is unthinkable for an advanced industrial society, such as France, to content itself with simply being a research center. We need to master the technologies that allow us to produce the advanced products which we conceptualize and design.

Many subsequent French innovation policy measures are grounded in this line of thought:

- **Legislation:** In a major step forward, in 1999 France enacted the Innovation and Research Act, which emphasized the creation of new firms that would transfer research results from public research organizations to French industry. Among other provisions, the law authorized government researchers to take part in innovative start-ups for **six years** without losing their civil service status, in effect incentivizing them to launch their own businesses.
- **Decentralization:** In 1992, the French government **began** moving 140 government research teams from Paris to 43 towns in the provinces, with the expectation that 4,500 researchers, engineers, and technicians would relocate to these areas by the end of the 1990s. Local governments were expected to support this effort by mobilizing regional scientific communities within the context of regional economic development plans.
- **Tax incentives:** The Research Tax Credit (CIR) established under Mitterrand was modified in phases between 2004 and 2008, eventually **underpinning** the "most attractive R&D tax regime in Europe." According to a 2021 study by Ernst and Young, over **77 percent** of French start-ups benefited from the CIR that year, which significantly helped cover their R&D outlays.
- **SME funding:** In 2005, consolidating several public organizations, France created the Oséo Group, an industrial and commercial institution facilitating funding for innovative SMEs on favorable terms. Oséo, which is **supported** in part by the European Investment Bank, emphasizes funding ventures in the start-up, innovation, development, and buy-out phases.
- **Clusters:** In 2005, France established a program promoting *pôles de compétitivité* ("**competitiveness clusters**"), public-private partnerships centered on specific geographic locations where companies and research organizations engage jointly in innovative projects. The government initially **designated** 67 clusters nationwide (possibly too many) and allocated 1.5 billion euros to support them in the years 2006–08 and another tranche of 1.5 billion euros for 2009–11. Designated clusters continue to receive government financial **support** through Oséo and the National Research Agency (ANR). Businesses belonging to clusters enjoy tax exemptions on R&D outlays and reduced social security contributions on behalf of R&D staff.
- **Carnot Institutes:** Starting in 2006, France began **designating** certain public applied research organizations as "Carnot Institutes," which represented not only a recognition of excellence but also qualification for "government funds correlated to the amount of revenue which each institute derives from contract research for French industry." This incentive-based system has

seen considerable success, perhaps because it has a mechanism to remove underperforming universities from the system.

- **Continuity:** Throughout the frequent modifications to French innovation policy, numerous changes in administrations, and the financial travails of certain French chip firms in a rapidly evolving global market, government support for the microelectronics sector has remained unwavering. Such policy continuity is also observable in other jurisdictions that have fostered competitive chip industries—examples include **Japan's** sustained support of its semiconductor industry during the Liberal Democratic Party's 38 years of uninterrupted governance post-World War II; **Taiwan's** strong support of its chip industry under both the Kuomintang and the Democratic Progressive Party; and **New York** state's decades-long promotion of the semiconductor regimes under a succession of Democratic and Republican governors. Sustained government support enables long-term planning, training, and investment, as well as commitment by individuals to an industry in which jobs are unlikely to be destabilized by changes in government.

The Cradle of French Microelectronics

The Electronics and Information Technology Laboratory (Leti) was created by the French government in 1967 within the CEA. Its original mission was to provide the French nuclear program with knowledge of automation, but by the early **1980s** it had "long since departed from that basic area and . . . [was] one of the most important chip laboratories in France."

CREATING NEW SEMICONDUCTOR COMPANIES

Soon after its creation, CEA-Leti began to focus on creating new semiconductor companies, some of which have become Europe's most globally competitive chip firms. Start-ups have been facilitated by a financial arm of the CEA, which invests an average of **500,000 euros** in each new venture to encourage further private sector investment. Initial progress was slow, but by 2022 CEA-Leti had supported more than **70 new companies**. The laboratory continues to collaborate with and support many of them, to remarkable effect:

- **STMicroelectronics**, Europe's largest semiconductor manufacturing and design firm, was "largely derived from Leti, where, after having manufactured a very first CMOS transistor in 1968, a subsidiary was created to industrialize the process, Efcis, which would give birth to SGS-Thomson, which became STMicro." In 1990, CEA-Leti transferred silicon-on-insulator technology to **Thomson-CSF**, and to **SGS-Thomson** (one of STMicro's corporate predecessors) it transferred the process technology for 16-bit erasable programmable read-only memory (EPROM) chips and a production line for 1.2-micron analog CMOS devices for telecommunications applications. CEA-Leti **supports** and sometimes creates SMEs in the Grenoble cluster, while its partner STMicro outsources the development of designs and components to the cluster's SMEs. Not all the firms have succeeded at scale, but today STMicro is regarded as one of Europe's most innovative firms, specializing in the production of analog chips and low-power sensors.
- Another example is **Soitec**, created in 1992 by four CEA-Leti researchers to use Smart-Cut™ technology to produce silicon-on-insulator (SOI) wafers on an industrial scale. Since **2018**, CEA-Leti has collaborated with Soitec on the Substrate Innovation Center (headed by a former

CEA-Leti director) to develop silicon carbide semiconductors, which are expected to play a key role in electric vehicles. Today, Soitec **boasts** that it is "one of the crown jewels of France's industrial sector." It has over 2,000 employees and **two** wafer production facilities, one in France and one in Singapore.

- Similarly, **Sofradir** was spun off from CEA-Leti in 1986 to exploit technology developed by the lab for infrared detectors. The company has continued to work with CEA-Leti, generating 55 patents as of 2016. Today, it is "the world's second-leading vendor of cooled infrared detectors for aerospace, defense, and industrial applications" and is "expanding its business to cover all technologies on semiconductor materials for infrared detection."

At present, CEA-Leti is accelerating its start-up activities. **According** to Director Sébastien Dauvé, "Our objective revolves around three to five creations per year, compared to two or three until now." Over time, Leti has an even more ambitious challenge to create deep tech start-ups based on disruptive technologies and backed by close long-term support to accelerate their technological development.

ENSURING LONG-TERM POLICY SUPPORT

French measures promoting innovation since 1982 have materially contributed to CEA-Leti's evolution and success. Pursuant to the government's creation of competitiveness clusters in 2005, the **Minalogic** digital electronics cluster, of which CEA-Leti was a founding member, was established in the Auvergne-Rhône-Alpes region. The cluster now has around 500 members, including 440 companies, and has **benefited** from a vast array of local, regional, national, and EU government programs. For example, CEA-Leti has been **designated** a Carnot Institute since the very inception of the program in 2006 and "proudly carries the Carnot label of scientific excellence and professionalism." As of 2014, CEA-Leti and the National Office for Aerospace Study and Research (ONERA) between them accounted for about **40 percent** of total Carnot funding, and other CEA-Leti spinoff start-ups have benefited **substantially** from the CIR R&D tax credit.

CEA-Leti was founded on the **premise** that it would derive "one industrial franc for one franc of subsidy." Over time, that ratio has evolved, with external research contracts now accounting for about 80 percent of its annual revenues (\$332.5 million in 2022). **According** to former CEA-Leti CEO Marie-Noëlle Semeria, "These are research contracts with industrialists and collaborative contracts at the national or European level. We only have 20 percent of recurring resources allocated by the State, which is very low for a research institute, but which allows us to explore and continue to take risks." While private sector investment is important—and validating—public support nonetheless remains vital. CEA-Leti receives **public funding** from over a dozen national, regional, and EU government organizations, including the European Commission, the French National Research Agency, the City of Grenoble, and the French Ministry of the Economy, Industry, and Digital Technology.

GROWING THE CEA-LETI COMPLEX: UNPARALLELED INFRASTRUCTURE

A major factor in CEA-Leti's growth is its ability to offer partners extraordinary **R&D infrastructure**. It has an on-site research staff of over 2,000 specialists in microelectronics disciplines, with half working on semiconductor processes such as deposition, etching, planarization, photolithography, and bonding and advanced materials and half working on architecture and applications. The facility itself boasts 11,000 square meters (118,400 square feet) of ISO-certified 9001 cleanroom space, which it aims to expand by 2,700 square meters (29,060 square feet) over the next few years. Importantly,

it operates experimental production facilities for numerous microelectronics applications, including 200- and 300-millimeter CMOS lines, 200-millimeter microelectronics machine (MEMS) lines, a 300-millimeter photonics platform, and a 300-millimeter 3D integration line. CEA-Leti also **operates** a nanocharacterization platform that examines materials properties for microelectronics applications and can draw upon cutting-edge equipment operated by Grenoble's European Synchrotron Radiation Facility (ESRF) and Institut Laue-Lanvin (ILL). Together, this represents an extraordinary collection of innovative research facilities.

RESEARCH ACTIVITIES

CEA-Leti's "lab-to-fab" approach entails deriving knowledge from basic research and shepherding it through various phases so it can ultimately be transferred to industrial partners. Progress has been remarkable. Recent research achievements include the development of a brain-computer interface, which enables thought-controlled walking in the wake of spinal cord injuries; a new chip architecture inspired by human synapses; and technology supporting sixth-generation (6G) wireless communications.¹

INTELLECTUAL PROPERTY

In what is possibly its most significant best practice, there is no single model for intellectual property (IP) management at CEA-Leti, with contractual arrangements tailored to the specifics of each project. In general, however, CEA-Leti retains ownership of the IP it develops, licensing this IP to its industrial partners. Reflecting these contributions, CEA-Leti currently holds over 4,000 patents.² This enables Leti to provide its partners with the knowledge required to advance their innovation projects, relying on a well-managed and continually enhanced intellectual property portfolio.

CREATING THE MINATEC INNOVATION CAMPUS

The Minatec innovation campus embodies French policymakers' long-sought objective of fusing research, education, and industrial innovation. Minatec is the brainchild of CEA-Leti's then-director, Jean Therme, who in 1999 conceived of a research complex in Grenoble that would combine public and private technological resources and know-how with academic programs in engineering to foster applied research partnerships in microelectronics and nanotechnology in a single location. Concerned that the deactivation of nuclear facilities in Grenoble would lead to local economic decline, Therme became a tireless advocate of a regional transition to microelectronics-based research. He secured institutional commitments and raised an initial sum of **150 million euros** from multiple **local authorities** (including the Department of Isère, the Auvergne-Rhône-Alpes region, and the City of Grenoble, which **offered** CEA-Leti an unoccupied piece of land downtown upon which to build).

Therme, who was dedicated to convening industry and academia to foster applied research, insisted on spending no more than two days per week in Paris, the epicenter of French basic science research. To ensure that representatives of different disciplines mingled with and engaged each other, the Minatec site featured only one canteen on site, and coffee machines were only located in communal areas.³

¹ CSIS roundtable with CEA-Leti, September 25, 2023.

² Ibid.

³ Peggy Hollinger, "Grenoble's Magnetic Force," *Financial Times*, June 24, 2005 (archived).

After several years of preparation, Minatec was launched in 2006 as a collaboration between CEA-Leti and the Grenoble Institute of Technology. It included a physical research center where industry partners could collaborate with technology transfer experts, basic research labs such as the Federation of Microelectronic and Nanotechnologies, and engineering schools. Today, the Minatec **campus** convenes 3,000 researchers, 600 technology transfer professionals, and 1,200 students.

FOCUSING ON LEGACY CHIPS

Reflecting the realities of global competition and European strengths, CEA-Leti emphasizes research designed to improve the energy efficiency of legacy, or foundational, chips needed by many European users, rather than seeking to compete with Taiwan and Korea in continuous node shrinkage or the pursuit of cutting-edge 2-nanometer manufacturing. CEA-Leti is **encouraging** EU-level policies that address the needs of Europe's leading automotive and telecommunications firms "by focusing on sensors, power electronics, and 5G telecom chips at less advanced nodes." As Director Sebastián Dauvé **said** in 2022:

Our position is to create value around electronics by mastering the basic technologies of microengraving at the level of silicon, sensors, radio frequency components or power electronics. Today, we are moving more toward so-called "more than Moore" functions, with varied applications for embedded intelligence in automobiles, health, industry, or the Internet of Things.

CEA-Leti has over **350** industrial partners, among them leading international chipmaking firms based in "like-minded countries."⁴ It has collaborated with many U.S. **companies**, including Qualcomm, IBM, Intel, Texas Instruments, Freescale (now part of NXP Semiconductors), and HP, in addition to other collaborations with like-minded countries.

FOSTERING COLLABORATIONS

Progress continues on promising new collaborative initiatives:

- **FD-SOI:** In 2022, CEA-Leti, Soitec, STMicro, and GlobalFoundries launched a joint effort to develop fully depleted silicon-on-insulator (FD-SOI) semiconductors. FD-SOI technology is **regarded** as an alternative to fin field-effect transistor (FinFET) technology for applications with strong cost, reliability, integration, or consumption constraints—such as artificial intelligence (AI) at the edge, the internet of things, automobiles, and high-frequency processors.
- **A new fab:** Concurrently, GlobalFoundries and STMicro **announced** plans in 2022 to build a 300-millimeter semiconductor plant in Crolles, about 13 miles from Grenoble, to develop FD-SOI technology, augmenting STMicro's existing 200- and 300-millimeter fabs at the site. This investment totals an **estimated** 7.5 billion euros, 2.9 billion of which was committed by the government of France.
- **EV chips:** In 2022, CEA-Leti entered into a **partnership** with Valeo, a French multinational maker of automotive components, to develop power electronics modules for applications in electronic vehicles (EVs).

⁴ CSIS roundtable with CEA-Leti, September 25, 2023.

- **Cutting-edge process technologies:** CEA-Leti is engaged in numerous process-technology collaborations with Intel, which is planning massive investments in chip production in Europe. These include a **project** to develop a die-to-wafer self-assembly process and a **project** aimed at placing atomically thin transition metal dichalcogenides (TMD) on 300-millimeter wafers to create nanosheet transistors. Additionally, in December 2023, CEA-Leti **unveiled** a new joint lab with Applied Materials to focus on materials engineering research for a variety of specialty applications.
- **Reducing energy consumption:** The **Leti 2030** program seeks to massively reduce the energy consumption of electronics by up to 1,000 times over the next decade. The organization forecasts that information and communication technologies, which today account for 14 percent of global electricity consumption, could account for only 4 percent by 2030.

ONGOING COOPERATION WITH THE UNITED STATES

Although CEA-Leti's presence in the United States has historically been limited, it collaborates in semiconductor research projects with several leading U.S. institutions:

- **Caltech:** CEA-Leti and the California Institute of Technology have **jointly developed** components for nanoelectromechanical systems (NEMS)—tiny devices like sensors, actuators (devices that cause machines to operate), and micromirror arrays that convert physical quantities such as speed, pressure, and direction into an electrical signal. In 2014, the partners transferred this technology to Apex Analytics, a Grenoble-based start-up that currently has 25 industrial partners in the United States, Europe, and Asia.
- **Stanford:** CEA-Leti and Stanford University have been collaborating on chip technology since 2016. In 2019, their research team **unveiled** a prototype for a "computer-on-a-chip," including processing circuits, memory storage, and power supply for applications such as monitoring crop moisture: "Equipped with machine learning algorithms, the chip could make on-the-spot decisions such as when to water. And with wireless technology it could send and receive data over the internet."
- **MIT:** The Massachusetts Institute of Technology has been collaborating with CEA-Leti since 2012 on the development of technologies such as chips that **facilitate** rapid tests for antibiotics, chips that **mimic** the working of the human brain, and semiconductor **nanowires** for solar cells, LEDs, and other applications from start-ups such as **Travera**.
- **CNSE:** CEA-Leti and the Albany-based College of Nanoscale Science and Engineering (CNSE) have been collaborating for 10 years with STMicro and IBM. Together, they have developed 14-nanometer FD-SOI transistors that use strain engineering (the process of tuning a material's properties by altering its mechanical or structural attributes). They have also used advanced transmission electron microscopy (TEM) to trace the evolution of polymer strains during the integration of stacked gate-all-around (GAA) nanosheet transistors.⁵

CEA-Leti states frankly that it is looking to deepen research collaborations with the United States, a "like-minded country," particularly with respect to lab-to-fab platforms and coordination on R&D

⁵ Ibid.

areas. Of particular shared interest are developing "chiplet" technology (the combination of advanced packaging and heterogeneous integration), applying advanced materials beyond CMOS, and matrix vector multiplication (MVM) operations for in-memory computing.⁶

Grenoble's High-Tech Cluster

The fact that a French version of Silicon Valley was emerging in Grenoble was evident by the early 1980s, when Dutch observers **noted**, "The essence of a real Silicon Valley lies in a large collection of small, rapidly growing, independent companies which constantly branch off into new ones. In Europe, only Grenoble comes anywhere near this kind of activity." When the French government launched its competitiveness cluster in 2005, Grenoble already enjoyed the government's "**unofficial blessing**" for cluster designation. Indeed, some have argued that "it was the city's pioneering efforts at uniting research, industry and education in the fields of micro and nanotechnology that **inspired** the government to launch the [cluster] campaign."

Even as the government was considering candidates for cluster designation, Philips, Freescale (later NXP Semiconductors), and STMicro were "investing \$3bn in a new research facility, Crolles 2, to develop high-performance technologies for advanced semiconductor chips"—**an alliance** that worked hand-in-hand with CEA-Leti. Between 1992 and 2007, around **\$4.8 billion** was invested in Grenoble's microelectronics and nanotechnology sectors. In part, this reflected the city's "strong tradition of public/private research partnerships"; by the time the region was designated a competitiveness cluster in 2005, approximately **70 percent** of the 200 patents filed by CEA-Leti each year were the result of industrial partnerships. The educational and training infrastructure supporting microelectronics activities in Grenoble were extraordinary even by global standards. As Freescale's advanced technology director, Andreas Wild, **said** in 2005, "We came because we were attracted by the competence in the area. . . . We studied 20 different sites before choosing Grenoble."

Today, Grenoble hosts the largest concentration of engineers in France and supports more than 25,000 research jobs, over 10,000 of which are in the private sector; the city also has the highest number of patents per capita in France at **8.5 per 10,000** inhabitants. Over 60,000 students attend its 10 engineering schools, including the elite Joseph Fourier University and Grenoble Institute of Technology. World-class high-tech companies have had a major presence in the area, **including** HP, Xerox, IBM, Orange, and STMicro. The density of the cluster **facilitates collaboration**, as do CEA-Leti's exceptional facilities: "CEA-Leti works with companies spread across the three valleys around Grenoble. One valley is a hotbed for microelectronics [the Grésivaudan Valley, where the city of Crolles is located], another is home to innovation in imagery systems, and the third valley is dedicated to display technologies."

Lessons from the CEA-Leti Model

CEA-Leti provides a powerful example of how to combine French statist traditions with strong connections to the private sector. The institution works closely with large companies, small companies, and start-ups. It draws on top-tier researchers, helping bring their skills and training to bear on both industrial challenges and societal problems. Its research facilities, university, and practical training

⁶ Ibid.

meet the needs of a cooperative research institution that is focused on both current industry needs and promising technologies.

In short, Leti's outstanding research facilities offer a major attraction, one that companies cannot provide themselves and that enables otherwise impossible technical advances.

Outstanding research facilities offer a major attraction, one that companies cannot provide themselves and that enables otherwise impossible technical advances.

From a public policy perspective, the continuity in the French approach has been essential. Successive governments have created an institution that attracts substantial resources for research, enabling it to create new and evolving programs while providing significant incentives for greater university-industry cooperation to address both commercial opportunities and social needs. This sizeable and sustained funding has proven to be a powerful engine for the cluster. Its exceptional research and educational facilities are complemented by a proven track record of successful international cooperation—which has, in turn, generated the trust to expand cooperation with new and growing programs in the United States and Europe. ■

***Sujai Shivakumar** is director and senior fellow of the Renewing American Innovation at the Center for Strategic and International Studies (CSIS) in Washington, D.C. **Charles Wessner** is a senior adviser (non-resident) with the CSIS Renewing American Innovation. **Thomas Howell** is an international trade attorney specializing in the semiconductor industry and a consultant with the CSIS Renewing American Innovation.*

This report is made possible by general support to CSIS. No direct sponsorship contributed to this report.

This report is produced by the Center for Strategic and International Studies (CSIS), a private, tax-exempt institution focusing on international public policy issues. Its research is nonpartisan and nonproprietary. CSIS does not take specific policy positions. Accordingly, all views, positions, and conclusions expressed in this publication should be understood to be solely those of the author(s).

© 2024 by the Center for Strategic and International Studies. All rights reserved.